



Database Forensics

How Data Solves Crimes in Large Organizations

**A 5 - slide guide to investigating data
at scale with SQL and audit logs**

EVODIA UNJUEH AKU
unjuehevo@gmail.com

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DATABASE FORENSICS = DIGITAL EVIDENCE

Collecting, analysing and reconstucting data from databases to investigate **fraud**, **breaches** and **anomalies** in large organizations.

Database

Organized collection of data that's stored and managed so you can quickly **find**, **update** and **analyze** it.

It's the difference between

- ❑ **The money is gone**
- and
- ❑ **The money left at 2:14 AM from user ID 4832 on IP 192.168.1.5.**

Forensics

Process of **collecting** and **analyzing** evidence to figure out **what happened**, usually after something went wrong.

Database forensics

Digging through a database logs and records to answer:

- ❑ **Who did what, when and how?**

SQL + Audit Logs + Timestamps = The Truth





WHY LARGE ORGANIZATIONS CAN'T IGNORE THIS

1. Scale = Complexity

Millions of transactions daily.

Manual checks
don't work.

Forensics uses SQL to find needles in haystacks.

2. Compliance and Legal Risk

GDPR, SOX, PCI.

Auditors demand proof. Database logs are the evidence trail.

3. Fraud Hides in the Data

70%+ of internal fraud shows up in db logs first.
Forensics catches it before it becomes a **headline.**





THE FORENSIC TOOLKIT

DATA SOURCES:

- ☐ Transaction logs
- ☐ Audit trails
- ☐ Backups

TOOLS:

- ☐ SQL and Python/Pandas
- ☐ SQL Server Profiler
- ☐ PgAudit

METHOD:

- ☐ Time-Based Analysis
- ☐ Anomaly Detection
- ☐ Evidence Reconstruction





TERMS CHEAT SHEET

- ❑ **Audit Trail:** Timestamped record of Who Changed What
- ❑ **Anomaly Detection:** Spotting unusual patterns in data
- ❑ **SOX:** US law requiring clean financial records
- ❑ **GDPR:** EU law requiring data privacy and audit trails

EXPLORE THE FULL REPO ON GITHUB.

